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Let $A \subset \mathbb{N}$ be such that every integer $n \geq 1$ can be written uniquely as $a_n - b_n$ for some $a_n, b_n \in A$. How fast must a_n/n increase?

#1194: [\[Er80,p.100\]](#)[additive combinatorics](#) | [additive basis](#) | [sidon sets](#)

The open status of this problem reflects the current belief of the owner of this website. There may be literature on this problem that I am unaware of, which may partially or completely solve the stated problem. Please do your own literature search before expending significant effort on solving this problem. If you find any relevant literature not mentioned here, please add this in a comment.

Comment activity that has not yet been incorporated into the remarks**None****Partial****Solution**

There are no solutions, partial or complete, claimed in the comments.

Such a set is called a perfect difference set; they can be constructed, for example, by a greedy approach (as detailed by Lev [\[Le04\]](#)). The greedy construction achieves $a_n \ll n^3$.

Erdős [\[Er80\]](#) wrote it is 'easy to see' that

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} = \infty.$$

Indeed, since A must be a Sidon set, if $a_n \leq f(n)n$ for all n , then all differences in $[1, x/f(x)]$ can be realised by differences from $A \cap [1, x]$, and hence $|A \cap [1, x]|^2 \gg x/f(x)$. Erdős (see for example [\[HaRo66\]](#)) proved that for infinitely many x $|A \cap [1, x]| \ll (x/\log x)^{1/2}$, which is a contradiction for a suitable choice of f .

This argument proves that $a_n \gg n \log n$ for infinitely many n .

Cilleruelo and Nathanson [\[CiNa08\]](#) describe a method for construction dense perfect difference

sets from dense Sidon sets, allowing many of the results for the latter to transfer to the former.

GPT-5.4 Pro (see the comments) has given an argument which proves that $a_n \gg n^{2-o(1)}$ infinitely often; in fact $a_n \gg n^2/f(n)$ infinitely often for any function f such that $\sum \frac{1}{nf(n)}$ diverges.

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Formalised statement? No ([Create a formalisation here](#))

Related OEIS sequences: Possible

6 comments on this problem

Likes this problem	None
Interested in collaborating	None
Currently working on this problem	None
This problem looks difficult	None
This problem looks tractable	None
The results on this problem could be formalisable	None
I am working on formalising the results on this problem	None

Additional thanks to: Liam Price

When referring to this problem, please use the original sources of Erdős. If you wish to acknowledge this website, the recommended citation format is:

T. F. Bloom, Erdős Problem #1194, <https://www.erdosproblems.com/1194>, accessed 2026-04-27

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